Arabic Post-OCR Correction with LLMs

Final Progress Report · 3 Experiments · 9 Datasets · 4 Models

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Primary model: Qwen3-4B-Instruct-2507 · OCR: Qaari (Qari-OCR-0.1-VL-2B)

Datasets: PATS-A01 (8 fonts) + KHATT + KHATT-Para + Yarmouk + Muharaf + Historical + BM + Kitab

Research Question & Motivation

Three experiments — progressive prompt engineering, model scaling, domain generalization

The Problem:

- Qaari (open-source Arabic OCR) produces systematic character-level errors:
- Dot-group confusions: ب/ت/ث/ي/ ن— identical skeletons, differ only in dots
- Similar-shape pairs: ف/ق, ر/ز, ص/ ض— easily confused under noise
- Runaway bug: Qaari repeats phrases on noisy/blank images (up to 17.7% of samples)

The Opportunity:

- LLMs have broad Arabic linguistic knowledge from pre-training
- Instruction-following enables prompt-based correction without fine-tuning
- Can post-OCR correction close the gap to expensive closed-source VLMs?

Evaluation methodology:

- Normal samples only (OCR/GT length ratio ≤5.0 — exclude Qaari runaways)
- Diacritics stripped before CER/WER — fair comparison across models
- Runaway corrector applied where noted (CER†): ratio threshold 3.0×GT

Experimental Structure

Exp.	Research Question	Runs
1 Phases 1–9	Does any LLM strategy improve Qaari OCR? Which knowledge type helps most?	8 phases
2 8 Trials	Which prompt design is optimal? Aggressive vs conservative vs RAG vs error-pattern?	8 trials
3 Final	Best prompt × 4 models × 2 OCR sources × 3 domains	9 runs

Primary metric throughout: CER (mean over normal samples, diacritics stripped)
Primary model: Qwen3-4B-Instruct-2507 | OCR: Qaari

Exp 1 Phased Knowledge Augmentation

8 experimental phases, each isolating one variable · PATS-A01 (8 fonts) + KHATT · Qwen3-4B

Phase 1: Qaari OCR baseline — error taxonomy, confusion matrix, runaway analysis

Phases 2–6,8,9: Progressive LLM correction — zero-shot → confusion matrix → self-reflective → combinations → RAG

Key finding: LLM correction is net harmful on PATS average; only RAG achieves below OCR baseline

Key finding: KHATT consistently improves ~1 pp CER across all strategies

Experiment 1 — Phase 1: Qaari OCR Baseline

Source: results_first_experiment_pats_khatt_only_8_phases_complete/phase1/

Font	CER	WER	N (normal)	Runaway %	Difficulty
Akhbar	2.12%	4.84%	437	4.8%	★ Easiest
Simplified	2.94%	5.81%	432	5.7%	★ Easiest
Traditional	3.63%	9.35%	435	5.0%	○ Medium
Arial	3.65%	6.84%	436	4.6%	○ Medium
Naskh	4.10%	9.56%	443	3.5%	○ Medium
Tahoma	5.82%	9.07%	407	10.3%	○ Medium
Thuluth	10.21%	31.11%	441	3.9%	✕ Hard
Andalus	12.12%	33.67%	427	6.2%	✕ Hard
PATS avg	5.57%	13.78%	3,458	5.5%	
KHATT	34.24%	75.60%	186	17.7%	✕ Hard

Key observations:

- PATS-A01 CER spans 2.12%–12.12% (5.7× range) across 8 fonts rendering identical text — typeface alone drives this variation
- KHATT handwritten: 34.24% CER = 6.1× harder than PATS average
- Runaway samples excluded from all LLM metrics (Qaari repetition bug)

Error taxonomy (PATS-A01 aggregate):

- 42.1% dot confusions (ب/ت/ث/ي/ن)
- 18.3% similar-shape substitutions (ف/ق, ر/ز)
- 11.6% hamza/alef variants
- 9.0% taa marbuta/ha confusions

KHATT error profile:

- 37% segmentation errors (word merges + splits) — rare in typewritten text
- Segmentation errors are structurally harder for text-only LLM correction

Two correction regimes:

- PATS: already accurate — LLM risks introducing more errors than it fixes
- KHATT: high noise — LLM has more signal but errors are harder to infer

** Normal samples only (OCR/GT ratio ≤5.0), diacritics stripped. CER† = after runaway correction (ratio threshold 3.0×GT).*

Experiment 1 — Phases 2–9: All LLM Strategies

PATS avg + KHATT | Δ vs Phase 1 OCR baseline | Red = harms PATS | Green = improves vs baseline

Phase	Strategy	PATS CER	Δ PATS	PATS WER	KHATT CER	Δ KHATT
P1	**OCR Baseline (Qaari)**	**5.57%**	**—**	**13.78%**	**34.24%**	**—**
P2	Zero-Shot LLM	5.84%	+0.27%↑	14.42%	33.25%	−0.99%↓
P3	OCR-Aware (confusion matrix)	5.74%	+0.17%↑	14.16%	33.26%	−0.98%↓
P4	Self-Reflective (training artifacts)	5.59%	+0.02%	13.66%	33.24%	−1.00%↓
P5†	CAMeL Validation (buggy†)	21.89%	n/a	90.74%	41.62%	n/a
P6	Combination: conf+self	5.76%	+0.19%↑	13.81%	33.27%	−0.97%↓
P8	RAG BM25 (char n-gram)	**5.45%**	**−0.12%↓**	12.04%	33.83%	−0.41%↓
P9	Error-Signature RAG	5.47%	−0.10%↓	13.63%	**33.13%**	−1.11%↓

† Phase 5 had implementation bugs (MorphAnalyzer init + position-keyed revert). Results are invalid; excluded from conclusions. | Best PATS: P8 RAG (5.45%, −0.12% vs baseline). Best KHATT: P9 Error-Sig (33.13%, −1.11%). All prompt strategies except RAG HARM or are neutral on PATS average.

Experiment 1 — Font-Level Impact: Which Strategies Help Which Fonts?

Green = improvement vs OCR baseline | Red = degradation | Values = CER% | Bold = best per font

Font	OCR	P2 ZS	P3 Conf.	P4 Self	P6 C+S	P8 RAG	P9 Err
Akhbar	2.12%	2.71%	2.57%	2.59%	2.46%	3.01%	2.27%
Simplified	2.94%	2.93%	2.97%	2.86%	2.72%	3.01%	2.56%
Traditional	3.63%	2.96%	2.90%	2.87%	2.75%	3.07%	2.88%
Arial	3.65%	4.50%	4.48%	4.08%	4.09%	3.83%	3.76%
Naskh	4.10%	4.27%	4.23%	4.22%	4.13%	4.36%	3.94%
Tahoma	5.82%	6.13%	6.12%	5.91%	6.15%	6.21%	5.76%
Thuluth	10.21%	10.92%	10.94%	10.67%	10.56%	9.68%	10.61%
Andalus	12.12%	12.27%	11.75%	11.51%	13.22%	10.41%	11.98%
PATS Avg	5.57%	**5.84%**	**5.74%**	**5.59%**	**5.76%**	**5.45%**	**5.47%**
KHATT	34.24%	33.25%	33.26%	33.24%	33.27%	33.83%	**33.13%**

P2 zero-shot harms 7 of 8 fonts (Traditional only exception: 3.63%→2.96%, −18.5%). P8 RAG is only strategy with PATS avg below OCR baseline. KHATT: all strategies improve by ~1 pp except P8 (slightly worse) and P9 (best: 33.13%).

** Normal samples only (OCR/GT ratio ≤5.0), diacritics stripped. CER† = after runaway correction (ratio threshold 3.0×GT).*

Experiment 1 — Key Findings

The over-correction problem dominates typewritten correction; RAG is the only reliable strategy

Finding 1 — Zero-shot LLM is net harmful on PATS:

- P2 zero-shot increases PATS avg CER from 5.57% to 5.84% (+4.8%)
- 7 of 8 PATS fonts are harmed; only Traditional improves (−18.5%)
- Root cause: model is "helpful" — rewrites correct tokens even with conservative prompt

Finding 2 — RAG achieves the only PATS improvement:

- P8 RAG BM25: 5.45% (−0.12% vs OCR baseline) — only strategy below baseline
- P9 Error-Sig RAG: 5.47% (−0.10%) — close second
- Domain-matched retrieval prevents false corrections; zero-shot examples on clean input ≈ no-op

Finding 3 — KHATT: modest but consistent improvement:

- All prompt strategies reduce KHATT CER by 0.4–1.1 pp
- P9 best: 33.13% (−1.11% abs vs OCR baseline 34.24%)
- Segmentation errors limit further gains (37% of KHATT errors = word merges/splits)

Finding 4 — No universal threshold:

- Traditional (3.63% CER) is helped; Arial (3.65%) is harmed
- Error type, not just baseline CER, determines whether LLM helps

Per-Font: Which Fonts Benefit from P8 RAG?

Font	OCR	P8 RAG	Δ	Verdict
Akhbar	2.12%	3.01%	+0.89%	✕ HARMED
Simplified	2.94%	3.01%	+0.07%	○ flat
Traditional	3.63%	3.07%	−0.56%	✓ helped
Arial	3.65%	3.83%	+0.18%	✕ harmed
Naskh	4.10%	4.36%	+0.26%	✕ harmed
Tahoma	5.82%	6.21%	+0.39%	✕ harmed
Thuluth	10.21%	9.68%	−0.53%	✓ helped
Andalus	12.12%	10.41%	−1.71%	✓ helped
PATS Avg	5.57%	5.45%	−0.12%	**✓ best**

Exp 2 Prompt Design Study — 8 Trials

Systematic prompt engineering: base vs conservative vs error-pattern × zero-shot vs RAG

13 datasets: PATS-A01 (8 fonts) + KHATT + KHATT-Para + Yarmouk + Muharaf + Historical
Model: Qwen3-4B-Instruct-2507 (same as Exp1)
Goal: Find the optimal prompt configuration for Experiment 3
Key finding: T3 (conservative zero-shot) best PATS (5.27%); T7 catastrophic (190%!)

Experiment 2 — 8 Prompt Trials: PATS & KHATT

13 datasets | normal-only, diacritics stripped | Qwen3-4B | ✓= best per column

#	Trial ID	Prompt Style	RAG?	PATS CER	PATS WER	KHATT CER	KHATT WER
OCR	**—**	**Baseline (no LLM)**	**—**	**5.57%**	**13.78%**	**34.24%**	**75.60%**
T1	base_zs	Base (aggressive)	No	5.84%	14.43%	33.25%	73.93%
T2	base_rag	Base	Yes	5.44%	**12.03%**	33.82%	73.76%
T3	cons_zs	Conservative	No	**5.27%**	13.35%	32.93%	74.15%
T4	cons_rag	Conservative	Yes	27.02%	28.10%	32.69%	73.29%
T5	hp_zs	Error-pattern	No	5.85%	13.95%	32.90%	72.78%
T6	hp_rag	Error-pattern	Yes	19.64%	24.52%	32.64%	73.40%
T7	hp_cons_zs	⚠ Err-pat+Cons	No	**190.60%**	198.12%	51.64%	90.58%
T8	hp_cons_rag	Err-pat+Cons	Yes	46.56%	48.79%	**32.54%**	73.67%

✓T3 (conservative zero-shot): best PATS — 5.27% (−5.4% relative vs OCR baseline). No RAG overhead. ✓T2 (base+RAG): best overall balance — 5.44% PATS + 33.82% KHATT. → Chosen for Experiment 3. ⚠ T7: catastrophic — prompt-template tags leaked into LLM output (template injection bug).

Experiment 2 — Per-Font PATS CER

Error-pattern prompts (T4, T6, T7, T8) cause catastrophic font-specific failures

Trial	Akhbar	Andalus	Arial	Naskh	Simplified	Tahoma	Thuluth	Trad.	Avg
OCR	2.12%	12.12%	3.65%	4.10%	2.94%	5.82%	10.21%	3.63%	5.57%
T1	2.71%	12.26%	4.50%	4.26%	2.94%	6.14%	10.93%	2.97%	5.84%
T2	2.97%	10.42%	3.83%	4.36%	3.01%	6.20%	9.68%	3.07%	5.44%
T3	2.32%	11.68%	3.74%	3.69%	2.43%	5.54%	10.23%	2.58%	**5.27%**
T4	25.32%	36.67%	22.7%	4.01%	2.56%	5.61%	**116.3%**	3.02%	27.0%
T5	2.83%	11.87%	4.24%	4.21%	3.01%	6.12%	11.26%	3.25%	5.85%
T6	4.02%	59.77%	5.28%	5.18%	38.96%	7.77%	10.22%	25.9%	19.6%
T7	**84.8%**	**153.0%**	**355.4%**	**300.1%**	**156.6%**	**73.9%**	**199.0%**	**202.0%**	**190.6%**
T8	3.25%	53.21%	4.21%	4.48%	196.4%	**6.43%**	81.4%	23.1%	46.6%

T3 (gold) beats T2 on 6/8 fonts. T2 wins Andalus (10.42% vs 11.68%) and Thuluth (9.68% vs 10.23%). Error-pattern prompts (T4-T8) are unreliable: different fonts fail unpredictably.

** Normal samples only (OCR/GT ratio ≤5.0), diacritics stripped. CER† = after runaway correction (ratio threshold 3.0×GT).*

Exp 3 Final Experiment — 3 Research Categories

Best prompt (T2 base_rag) · 4 models · 2 OCR sources · BM Bench + Kitab

3A — Model comparison: 4 models × val set × Qaari OCR × T2 (9 datasets each)

3B — OCR source impact: Qwen3-4B × val set × T2 × Qaari vs Gemma-3 VLM as OCR engine

3C — Generalization: Qwen3-4B × T2 × RDI-Test Benchmark + Kitab × 2 OCR sources

Runaway corrector applied (CER†): critical for Qwen3-14B which has massive runaway epidemic

Experiment 3A — Model Comparison

val set · Qaari OCR · T2 (base_rag) | CER = raw output | CER† = after runaway correction | FP = Full-page group

Model	Params	PATS CER	PATS CER†	KHATT CER	KHATT CER†	FP CER†
OCR Baseline	—	**5.57%**	—	**34.24%**	—	**86.50%**
Qwen3-4B	4B	5.44%	5.44%	33.82%	**33.82%**	**68.18%**
Qwen3-14B	14B	**15.30%**	5.60%	**175.69%**	35.55%	88.51%
Gemma-3-4B	4B	7.47%	7.47%	36.14%	36.14%	71.12%
Gemma-3-12B	12B	5.33%	**5.33%**	35.98%	35.98%	70.01%

Per-font PATS CER† (runaway-corrected)

Font	OCR	Q3-4B	Q3-14B	G3-4B	G3-12B
Akhbar	2.12%	2.97%	3.27%	4.80%	3.05%
Simplified	2.94%	3.01%	3.39%	4.93%	3.26%
Traditional	3.63%	3.07%	3.96%	5.48%	3.35%
Arial	3.65%	3.83%	4.50%	5.90%	3.67%
Naskh	4.10%	4.36%	4.75%	5.94%	4.41%
Tahoma	5.82%	6.20%	6.10%	7.64%	6.03%
Thuluth	10.21%	9.68%	9.06%	11.81%	8.86%
Andalus	12.12%	10.42%	9.80%	13.26%	10.00%
Avg	5.57%	5.44%	5.60%	7.47%	**5.33%**

Key: Runaway corrector essential for Q3-14B (KHATT 175.69%→35.55% with fix). G3-12B best PATS (5.33%); Q3-4B best KHATT (33.82%) + FP (68.18%). FP = Full-page: OCR 86.50% → Q3-4B 68.18% (−21% abs). G3-4B consistently worst; larger scale helps only within each family.

** Normal samples only (OCR/GT ratio ≤5.0), diacritics stripped. CER† = after runaway correction (ratio threshold 3.0×GT).*

Experiment 3B — OCR Source Quality Impact

Qwen3-4B · T2 | Full-page datasets only (Gemma excluded for line-strip images)

⚠ Gemma-3 VLM OCR is EXCLUDED for PATS and KHATT (line-strip images, 10–16:1 aspect ratio). Gemma's image preprocessor squashes these to near-square → hallucinations + loops → results invalid. Gemma results reported only for full-page datasets (~0.6:1 ratio) where the comparison is valid.

Full-page datasets — Qaari vs Gemma OCR (valid comparison)

Dataset	OCR Src	OCR CER	OCR WER	Corr CER†	Corr WER†
KHATT-Paragraph-val	Qaari	61.68%	91.24%	45.06%	68.49%
	Gemma	36.14%	64.43%	35.82%	63.82%
Yarmouk-testing	Qaari	49.85%	78.32%	43.35%	69.94%
	Gemma	95.76%	140.20%	74.37%	111.74%
Muharaf-validation	Qaari	129.39%	152.67%	89.78%	123.10%
	Gemma	141.15%	182.26%	72.85%	109.99%
Historical	Qaari	105.10%	136.81%	94.53%	127.03%
	Gemma	76.26%	121.77%	74.09%	118.52%
Full-page Avg	**Qaari**	**86.50%**	**116.67%**	**68.18%**	**80.60%**
Full-page Avg	Gemma	87.33%	123.17%	**64.28%**	88.05%

Key findings:

Full-page comparison:

- Gemma OCR ≈ Qaari (87% vs 87%)
- Gemma corrected: 64.28% vs Qaari 68.18%
- LLM extracts more from Gemma FP output

Why not PATS/KHATT:

- Line strips: 10–16:1 ratio
- Gemma preprocessor: squashes to square
- Result: hallucinations + loops → invalid

Implication:

- OCR source matters less for full-page
- Line-strip OCR: Qaari is the reliable engine
- VLM OCR needs aspect-ratio-aware preprocessing

** Normal samples only (OCR/GT ratio ≤5.0), diacritics stripped. CER† = after runaway correction (ratio threshold 3.0×GT).*

Experiment 3C — Generalization to New Domains

Qwen3-4B · T2 · British Museum Arabic Benchmark + Kitab Benchmark | CER†

RDI-Test-Lines (line-strip images, text GT)

Note: RDI-Test Line Segmentation excluded — polygon-only GT, no text transcription.

Subset	OCR Src	N	OCR CER	Corr CER†	Δ
LR-Handwritten	Qaari	385	98.82%	91.55%	−7.3%
	Gemma	762	62.53%	57.71%	−4.8%
LR-Manuscripts	Qaari	1009	81.94%	79.00%	−2.9%
	Gemma	1175	78.10%	70.36%	−7.7%
LR-Typewritten	Qaari	671	105.19%	84.79%	**−20.4%**
	Gemma	1314	64.86%	55.31%	−9.5%
RDI-Test-Lines Overall	**Qaari**	**2065**	**95.31%**	**85.11%**	**−10.7%**
RDI-Test-Lines Overall	Gemma	3251	68.49%	61.13%	−10.7%

Kitab Benchmark (13 subsets)

Subset	OCR Src	OCR CER	Corr CER†	Δ
kitab-hindawi	Qaari	31.46%	24.31%	−7.2%
kitab-historyar	Qaari	63.48%	51.69%	−11.8%
kitab-muharaf	Qaari	76.04%	58.72%	−17.3%
kitab-isippt	Gemma	87.48%	68.75%	−18.7%
kitab-patsocr	Gemma	87.98%	69.95%	−18.0%
kitab-khattpar	Qaari	205.27%	**90.60%**	−114.7%
kitab-arabicocr	Qaari	1.80%	7.21%	+5.4%⚠
kitab-evarest	Qaari	32.92%	60.67%	+27.8%⚠
Kitab Overall	**Qaari**	**49.28%**	**40.73%**	**−17.3%**
Kitab Overall	Gemma	79.45%	60.07%	**−24.4%**

RDI-Test-Lines: Gemma OCR better than Qaari on Handwritten (62% vs 98%) and Typewritten (65% vs 105%) — Qaari struggles on diverse BM images. Kitab −17% (Qaari) / −24% (Gemma). Low-CER subsets harmed (arabicocr 1.8%); khattparagraph 205%→90% (runaway corrector essential).

** Normal samples only (OCR/GT ratio ≤5.0), diacritics stripped. CER† = after runaway correction (ratio threshold 3.0×GT).*

Summary — Final Results & Conclusions

All metrics: normal samples only, diacritics stripped, CER† = runaway-corrected

Research Answer: Yes, with important caveats

Exp1 — Phased strategies:

- LLM correction net harmful on PATS avg (zero-shot: +4.8%)
- Only RAG achieves below OCR baseline: P8 5.45% (−0.12%)
- KHATT: ~1 pp improvement across all strategies (P9 best: 33.13%)
- Over-correction: 7/8 fonts harmed by zero-shot; font-specific, not CER-threshold-based

Exp2 — Prompt design:

- Conservative zero-shot (T3): best PATS 5.27% (−5.4% relative)
- Base+RAG (T2): best overall balance — used in Exp3
- Error-pattern prompts: risky — catastrophic on some fonts

Exp3 — Final results:

- Best PATS: Gemma-3-12B 5.33% | Best KHATT: Qwen3-4B 33.82%
- Qwen3-14B: runaway epidemic (175%→35% with corrector)
- Full-page: OCR 86.50% → 68.18% (−21% abs)
- Generalizes: BM −10.7%, Kitab −17–24%
- Gemma VLM OCR: excluded for line strips (invalid); better than Qaari on RDI-Test-Lines images

Best Results at a Glance

Task / Dataset	Best Config	CER	vs OCR
PATS (8 fonts)	G3-12B, T2, Qaari	**5.33%**	−4.3% rel
PATS (no RAG)	T3 cons_zs, Q3-4B	**5.27%**	−5.4% rel
KHATT	Q3-4B, T2, Qaari	33.82%	−1.2% rel
Full-page	Q3-4B, T2, Qaari	68.18%	−21% abs
RDI-Test Benchmark	Q3-4B, T2, Qaari	85.11%	−10.7% abs
Kitab (Qaari)	Q3-4B, T2, Qaari	40.73%	−17.3% rel
Kitab (Gemma OCR)	Q3-4B, T2, Gemma	60.07%	−24.4% rel